Activation Functions in Neural Networks - ReLU vs Sigmoid vs Tanh

1. Introduction

Models based on Artificial Neural Networks have come to be one of the commonly used machine learning models for solving complex pattern recognition problems like image classification, speech recognition, and natural language processing. The layered networks of artificial neurons known as neural networks learn to represent the data with backpropagation.

The activation function is an important part of a neural network, as it determines how the output of a neuron is transformed before passing it on to the next layer. Neural networks would behave like simple linear models without activation functions and would not be able to learn complex relationships in data.

Activation functions help in introducing non-linearity into the neural network making them learn complex patterns. The gradient propagation throughout the network, model learning speed, and generalization to unseen data is dependent on the type of activation function.

This tutorial investigates the behaviour of three commonly used activation functions:

- Sigmoid
- Tanh
- ReLU (Rectified Linear Unit)

The objective of this tutorial is to compare these activation functions in terms of training behaviour and classification performance using the MNIST handwritten digit dataset. By analysing learning curves, loss behaviour and classification accuracy, the tutorial demonstrates how activation function choice influences neural network performance.

2. Activation Functions in Neural Networks

Activation functions determine the output of each neuron in a neural network. They introduce non-linear transformations that allow neural networks to approximate complex functions.

Sigmoid Function:

The sigmoid activation function transforms input values into a range between 0 and 1.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Early neural networks often made use of sigmoid functions since their output values are like probabilities. Despite this, the gradient of the sigmoid function becomes small during backpropagation. The effect is a slowdown in learning in deeper networks.

Tanh Function:

The hyperbolic tangent function (tanh) is like the sigmoid function but maps values to a range between −1 and 1.

$$tanh(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

Because the tanh function is zero-centred, it often performs better than sigmoid in practice. However, it still suffers from vanishing gradients for large positive or negative inputs.

ReLU Function:

The Rectified Linear Unit (ReLU) is currently one of the most widely used activation functions in deep learning.

$$ReLU(x) = max(0, x)$$

ReLU is computationally efficient and reduces the vanishing gradient problem. As a result, ReLU often enables faster and more stable training in deep neural networks.

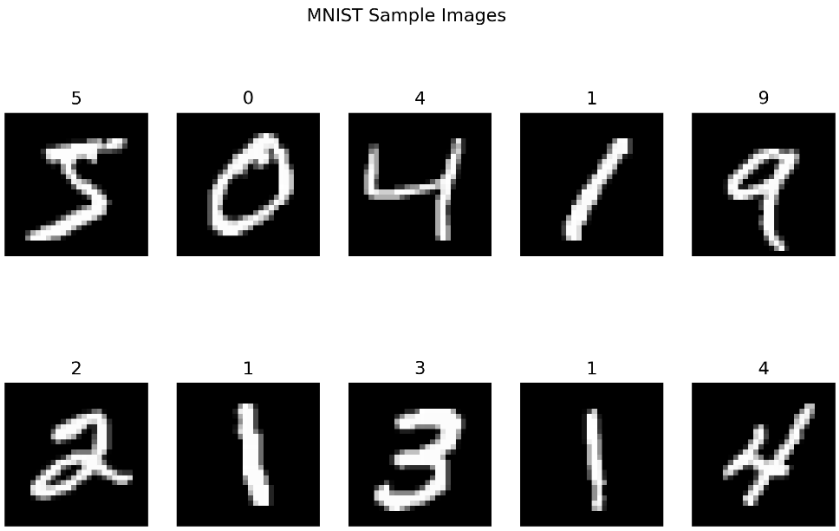
3. Dataset

This tutorial uses the **MNIST dataset**, one of the most widely used benchmark datasets in machine learning research (LeCun et al., 1998). The dataset contains grayscale images of handwritten digits and is commonly used for evaluating classification algorithms.

Each image has a resolution of 28 × 28 pixels and represents a handwritten digit between 0 and 9.

The MNIST dataset is particularly suitable for this tutorial because it provides a simple image classification task while still allowing neural networks to demonstrate different learning behaviours depending on the chosen activation function.

Before training the models, the pixel values were normalised to the range 0–1 to improve training stability and optimisation efficiency.



4. Experimental Setup

To analyse the impact of activation functions, three neural network models were trained using identical architectures but different activation functions.

The neural network architecture used in this experiment consists of:

- Input layer (flattened 28 × 28 image)
- Two hidden layers with 128 neurons each
- Output layer with 10 neurons representing digit classes.

The hidden layers use one of the following activation functions:

- ReLU
- Sigmoid
- Tanh

Parameter	Value

Optimizer	Adam
Loss function	Sparse categorical crossentropy
Epochs	10
Batch size	128
Validation split	20%

During training, several visualisations were generated to compare model behaviour, including:

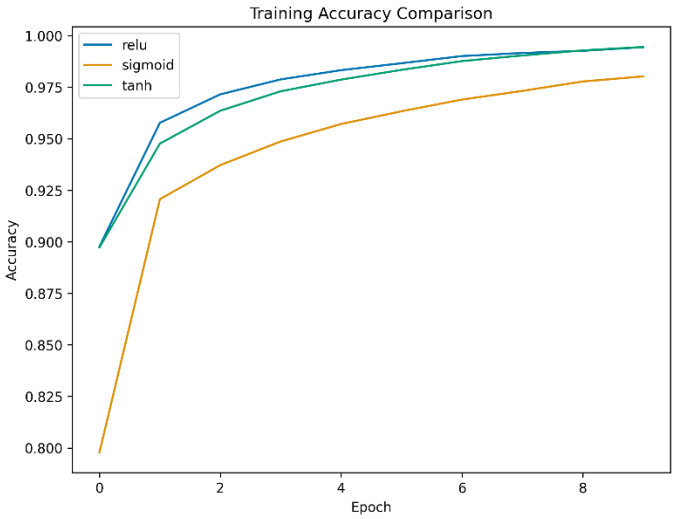
- training accuracy curves
- validation accuracy curves
- loss curves
- final classification accuracy comparison
- confusion matrix analysis
- prediction examples

These visualisations help illustrate how activation functions influence neural network training dynamics.

5. Results and Analysis

5.1 Training Accuracy Comparison:

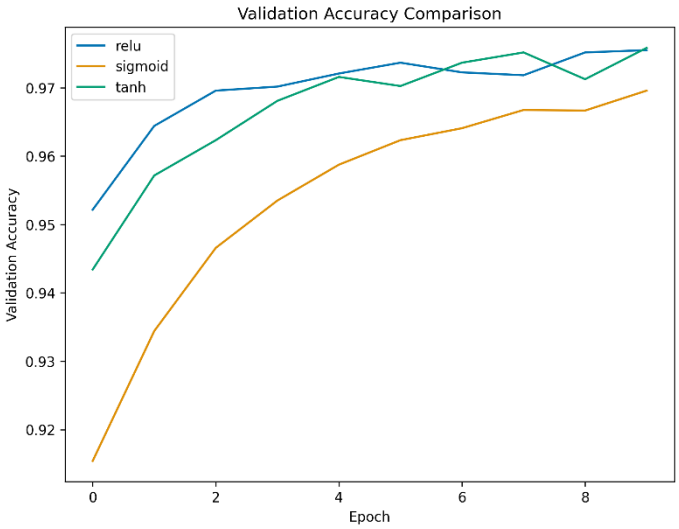
Training accuracy was analysed to observe how quickly each activation function enables the model to learn.



The results show that the ReLU activation function typically allows the model to achieve higher training accuracy faster than sigmoid or tanh. This is due to ReLU's ability to avoid vanishing gradients and maintain stronger gradient signals during training.

5.2 Validation Accuracy Comparison:

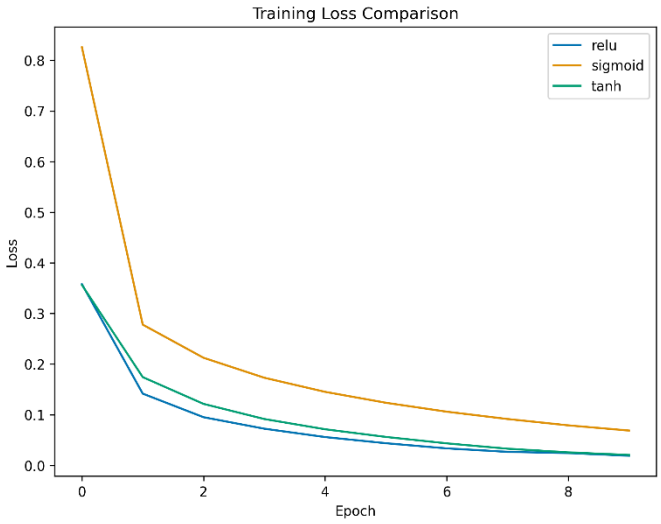
Validation accuracy provides insight into how well the model generalises to the unseen data.



The validation accuracy curves demonstrate that ReLU often provides more stable learning behaviour. Sigmoid and tanh models tend to converge more slowly and may experience slower improvements in accuracy.

5.3 Training Loss Behaviour:

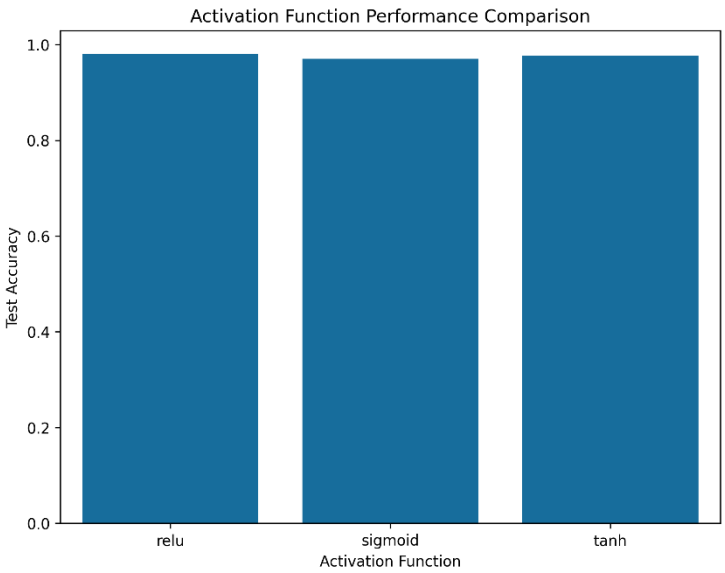
Training loss indicates how efficiently the neural network minimises classification error during training.



The loss curves show that ReLU networks reduce loss more quickly than sigmoid-based networks. This behaviour highlights the advantage of ReLU in maintaining stronger gradient signals.

5.4 Final Accuracy Comparison:

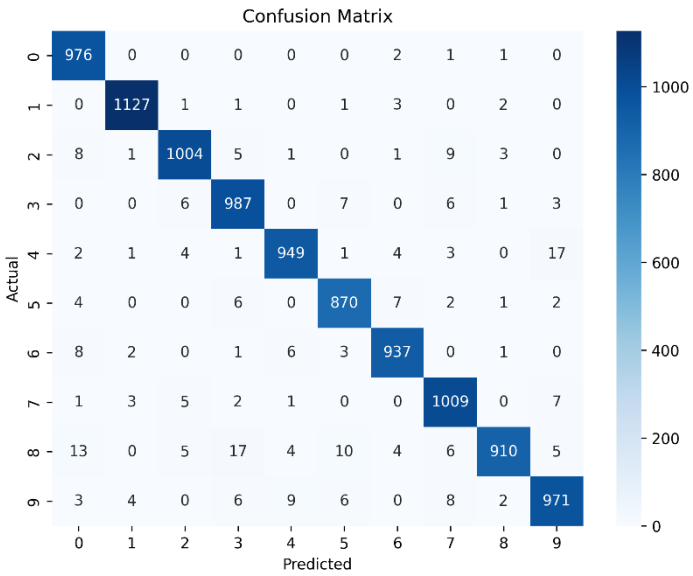
To compare the overall effectiveness of each activation function, final test accuracy was evaluated.



The results indicate that ReLU typically achieves the highest classification accuracy on the MNIST dataset. Tanh often performs moderately well, while sigmoid networks tend to learn more slowly.

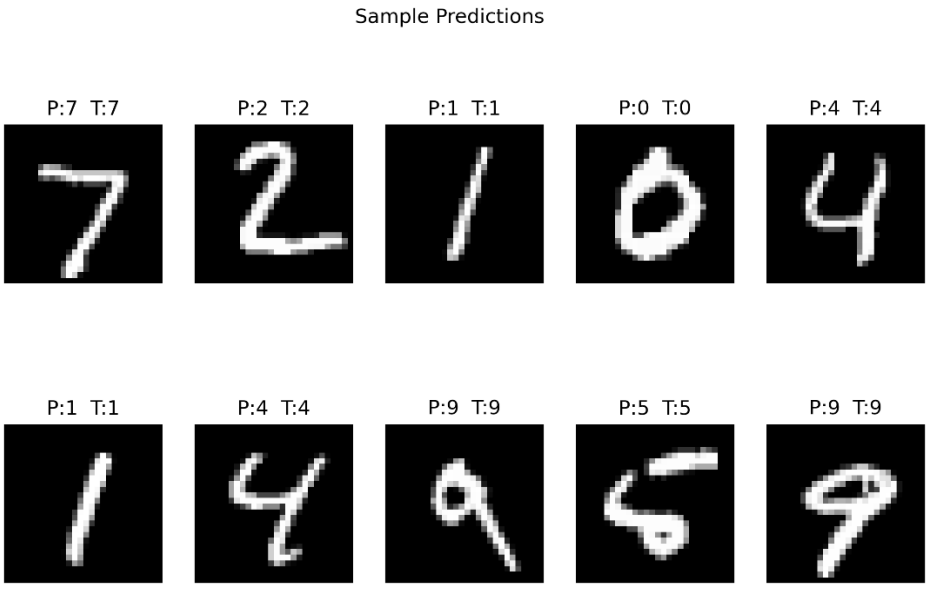
5.5 Confusion Matrix Analysis:

A confusion matrix was generated for the best performing model to analyse classification behaviour in more detail.



The confusion matrix shows that most digits are classified correctly, although some visually similar digits such as 4 and 9 may occasionally be confused.

5.6 Example Predictions:
Example predictions provide an intuitive demonstration of the model's classification performance.



6. Discussion

Experimental evaluation reveals the key role that activation functions play in the learning behaviour and performance of neural networks. All three neural networks used in this experiment are identical in architecture and training setup but differ in the choice of activation function, which results in different training behaviour and final classification accuracy.

A key observation from the results is that the ReLU activation function allows for faster convergence in training compared to the sigmoid and tanh activation functions. The accuracy curves for the training show that ReLU based networks achieve high accuracy in early epochs. This happens because the ReLU function has a stronger gradient on the backpropagation. Because the ReLU function has a constant gradient for all positive inputs unlike the sigmoid and tanh functions, which can have small gradients for inputs which fall into the saturated region. Thus, gradient updates stay effective even for deeper networks.

On the other hand, learning behaviour of sigmoid activation function is slower. The known vanishing gradient problem can explain this. When the input of the sigmoid function is large or small. The output will be either 1 or 0 (either close to 1 or 0). The gradient in this case becomes small. As the gradient travels through many layers during backpropagation, weight adjustments become extremely small. As a result, the network learns more slowly and may not be able to improve its accuracy.

The tanh activation function frequently outperforms the sigmoid function, as reflected in multiple studies of neural networks. One improvement is the fact that the tanh function is zero-centred, i.e. its output values all lie between −1 and 1. Zero-centred activations can improve optimisation since they allow the gradients to propagate more symmetrically through the network. But still, tanh function suffers with gradient saturation with large values of input.

A notable finding from the experiment is the difference in training stability associated with activation functions. According to the validation accuracy curves, Relu usually brings more stable improvements during training. During the training phase, sigmoid and tanh models might experience a sluggish rise in validation accuracy or fail to stabilise it early on. According to the study, it is easier to optimize networks that use ReLU with gradient-based learning algorithms.

The classification performance is better explained with the confusion matrix results. Most of the digits are correctly classified in all models which means even simple neural networks perform well on the MNIST data. Despite the presence of certain digits that are similar – like 4 and 9, or 3 and 5 – which can cause confusion. The incorrect placements show how difficult handwritten digits can get especially when written in an ambiguous form.

The MNIST dataset's relative simplicity is another source that may drive this result Due to the simple patterns and small size of MNIST images, even shallow neural networks can achieve high accuracy. Some activation functions may perform better on complex datasets that contain richer visual structures, it has been observed. For example, deeper architectures trained on complex datasets tend to benefit very much from ReLU because it prevents gradient flow from vanishing during training.

From a practical point of view, this experiment reinforces why ReLU is now the go-to activation function in deep learning. The computational simplicity, vanishing gradient effect mitigation, deep network compatibility, and other features recommend it for large scale machine learning tasks. While the sigmoid and tanh functions can be useful in some contexts, e.g. in the outputs (to reconstruct probabilities) or in some recurrent architectures, they are less commonly used as hidden layer activations now.

Overall, the outcomes of this experiment indicate that the choice of activation functions ought not to be considered a trivial implementation detail. This is a non-trivial design choice that can significantly affect training efficiency, gradient flow, and the final model accuracy. Taking careful consideration of activation functions can lead to an improvement in performance for neural networks and in model training.

7. Conclusion

- Activation functions significantly influence neural network learning behaviour.
- ReLU enables faster training and higher accuracy compared to sigmoid.
- Tanh performs better than sigmoid but still suffers from gradient saturation.
- Sigmoid activation functions are less effective in deeper neural networks.
- Choosing an appropriate activation function improves training efficiency and model performance.

Overall, the results highlight the importance of selecting activation functions carefully when designing neural network architectures.

8. Future Work

- Investigating modern activation functions such as **Leaky ReLU, ELU and Swish**.
- Evaluating activation function performance on **more complex datasets such as CIFAR-10**.
- Exploring the impact of activation functions in **deeper neural networks**.

9. References

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